THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In the pattern $4, 11, 18, 25, \ldots$ each term is 7 more than the one before. The 12th term is: 1

(a) 95 **(b)** 84 **(c)** 81 **(d)** 88

2. What is the angle between the hands of a clock at 4 o'clock, measured the long way round (the larger angle)? 1

(a) 210 **(b)** 120 **(c)** 240 **(d)** 270

3. The smallest EVEN number that can be formed by using each of the digits 3, 1, 4, 2 once is: 1

(a) 4312 **(b)** 1342 **(c)** 1324 **(d)** 1234

4. The total of the two largest values in the table is: 1

Item	Number
Apple	10
Banana	70
Orange	30

(a) 40 **(b)** 105 **(c)** 80 **(d)** 100

5. The number of factors of 48 is: 1

(a) 10 **(b)** 5 **(c)** 12 **(d)** 4

6. A rectangle has perimeter 50 cm and length 18 cm. Its area (in cm^2) is: 1

(a) 144 **(b)** 133 **(c)** 126 **(d)** 108

7. $5\frac{1}{2} - 2\frac{1}{3} =$ 1

(a) $3\frac{1}{2}$ **(b)** $3\frac{1}{6}$ **(c)** 4 **(d)** $3\frac{5}{6}$

8. A circle has centre O and radius 6 cm. Points are at distances 3, 6, 8, 5, 10 cm from O. How many of them lie inside the circle (not on it)? 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

9. A regular polygon has each angle of rotation $20°$. The number of lines of symmetry is: 1

(a) 20 **(b)** 17 **(c)** 18 **(d)** 19

10. -15 + 8 + (-6) = 1

(a) -11 **(b)** -3 **(c)** -13 **(d)** -15

11. In the pattern $3, 8, 13, 18, \ldots$ which term number equals 88? 1

(a) 17 **(b)** 19 **(c)** 18 **(d)** 23

12. The minute hand starts at 12 and stops at 4. The angle it turns through (clockwise) is: 1

(a) 90 **(b)** 240 **(c)** 120 **(d)** 150

13. The number of 3-digit palindromes is: 1

(a) 81 **(b)** 99 **(c)** 100 **(d)** 90

14. A pictograph has one symbol = 15. The number of symbols for A is 9 and for B is 4. How many more does A have than B? 1

(a) 90 **(b)** 5 **(c)** 75 **(d)** 195

15. The HCF of 36 and 48 is: 1

(a) 12 **(b)** 36 **(c)** 24 **(d)** 144

16. The area of a square is 36 cm^2. If its side is doubled, the new area (in cm^2) is: 1

(a) 144 **(b)** 156 **(c)** 138 **(d)** 150

17. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} =$ 1

(a) $\frac{5}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{3}{11}$ **(d)** 1

18. A square has a diagonal that splits it into two triangles. If the side is 5 cm, the sum of the angles of one triangle is (degrees): 1

(a) 540 **(b)** 90 **(c)** 270 **(d)** 180

19. The total angle through which a regular hexagon turns when rotated 5 times through its smallest angle (degrees): 1

(a) 300 **(b)** 310 **(c)** 290 **(d)** 330

20. 10 - (-4) - 6 = 1

(a) 6 **(b)** 10 **(c)** 8 **(d)** 18

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. In the pattern $5, 8, 11, 14, \ldots$ state the rule for getting any term from the first term, and find the 20th term. 2

22. Find the smaller and the larger angle between the hands of a clock at 5 o'clock, and name their types. 2

23. Find the greatest and the smallest 5-digit numbers formed from 4, 9, 1, 7, 3, each digit once. 2

24. Find the difference between the highest and lowest and the total of all. 2

Item	Number
Apple	75
Mango	70
Orange	35

25. Find the HCF and LCM of 24 and 36 using prime factorisation. 2

26. The perimeter of a rectangle is 70 cm and its length is 24 cm. Find the breadth and the area. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A row of triangles is made with sticks: 3 sticks for the first triangle and 2 more for each new triangle. Find the sticks needed for 12 triangles, and the number of triangles that can be made with 35 sticks. 3

28. A ray OA is turned about O through a three-quarter turn clockwise and then through a quarter turn anticlockwise. Find the net turn in degrees and its type. 3

29. Perform two Kaprekar steps on 1795. 3

30. Find the fraction of students liking Kabaddi and the number who do not like it. 3

Sport	Students
Kabaddi	25
Chess	40
Tennis	65

31. Find the prime factorisation of 180 and the number of its factors. 3

OR

A rectangular lawn 30 m by 20 m has a path 2 m wide inside it all around. Find the area of the path. 3

32. A farmer ploughed $\frac{1}{4}$ of a field on Monday and $\frac{1}{3}$ of it on Tuesday. What part of the field remains to be ploughed? 3

OR

A square of side 6 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A pattern of squares: the first figure has 5 squares and each next figure has 4 more. Find the number of squares in the 10th figure, and the first figure that has more than 91 squares. 5

34. A clock shows 1:00. The hour hand turns clockwise until the clock shows 3:00. Find the angle turned by the hour hand, the angle turned by the minute hand in this time, and the smaller angle between the hands at 3:00. 5

35. A 4-digit palindrome has first digit 2 and digit sum 10. Find the number. Add 11 to it and see if the answer is a palindrome. 5

36. A pictograph (one symbol = 12) shows 3 symbols on Monday, 6 on Tuesday and 9 on Wednesday. On Thursday the number of items sold equals the total of Monday and Tuesday. Find the number sold on each day, and the total for the four days. 5

OR

A room is 450 cm long and 360 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

37. A rectangular hall 10 m by 8 m is to be covered with square tiles of side 50 cm costing ₹40 each. Find the number of tiles and the total cost. 5

OR

Ravi had ₹120. He spent $\frac{1}{4}$ of it on books and $\frac{1}{3}$ of the rest on a bag. How much does he have left? 5

38. Write the steps to construct a square ABCD of side 4 cm using only ruler and compass. If the full diagonal is 6 cm, state the half-diagonal OA and the angle formed by the diagonals. 5

OR

A rangoli is a regular 5-pointed pattern. Find 5

(a) its lines of symmetry,

(b) the smallest rotation angle,

(c) how many rotations of that angle bring it back to the start,

(d) whether a rotation of 144° leaves it unchanged.

END OF QUESTION PAPER **Total: 80 Marks**

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